Fundamentals of Tensor Computation

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What is a tensor?

- A tensor is just a generalization of scalars, vectors, and matrices.
- A scalar is a 0th-order Tensor, a vector is called order 1st-order Tensor, a matrix is a 2nd-order Tensor, while a 3D collection of numbers is a 3rd-order Tensor.

Order	Example	Terminology
0	1	Scalar
1	$\begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$	Vector
2	$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}$	Matrix
3	$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}$	3rd-order tensor

A third order tensor

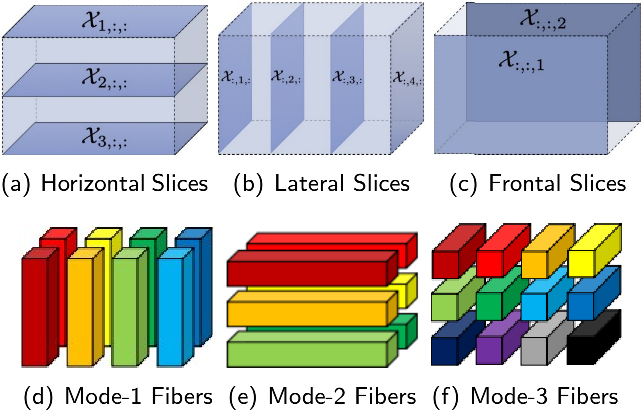
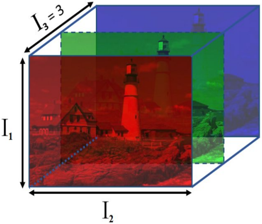
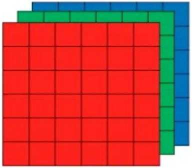


Figure: Illustration of a $3 \times 4 \times 2$ order tensor

[0] Source: https://ibug.doc.ic.ac.uk/media/uploads/documents/tensor_methods_in_computer_vision_and_deep_learning.pdf

Examples of 3rd and 4th order tensors





Color image is a 3rd-order Tensor

Color video is a 4th-order Tensor

[0] Sources: https://www.researchgate.net/publication/307091456_Quantifying_Blur_in_Color_Images_using_Higher_Order_Singular_Values, https://www.slideshare.net/BertonEarnshaw/a-brief-survey-of-tensors

Algebraic operations

- k-mode product
- Tensor-tensor product
  - t-product
  - M-product
- Einstein product

k-mode product[1] (tensor-matrix)

Let $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_k \times \cdots \times I_N}$ and $B \in \mathbb{R}^{J \times I_k}$. Then, their *k*-mode product is a tensor in $\mathbb{R}^{I_1 \times \cdots \times I_{k-1} \times J \times I_{k+1} \times \cdots \times I_N}$

$$(\mathcal{A} \times_k B)_{i_1 i_2 \ldots i_{k-1} j i_{k+1} \ldots i_N} = \sum_{i_k=1}^{I_k} a_{i_1 i_2 \ldots i_{k-1} i_k i_{k+1} \ldots i_N} b_{j i_k}.$$

[1] Lathauwer, L. D.; Moor, B. D.; Vandewalle, J., SIAM J. Matrix Anal. Appl. 21 (2000) 1253-1278.

Third-order tensor product

Let $\mathcal{A} \in \mathbb{F}^{I \times J \times K}$, then

$$\operatorname{bcirc}(\mathcal{A}) = \begin{bmatrix} \mathcal{A}^{(1)} & \mathcal{A}^{(K)} & \dots & \mathcal{A}^{(2)} \\ \mathcal{A}^{(2)} & \mathcal{A}^{(1)} & \dots & \mathcal{A}^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{A}^{(K)} & \mathcal{A}^{(K-1)} & \dots & \mathcal{A}^{(1)} \end{bmatrix} \in \mathbb{F}^{IK \times JK}.$$

[1] Kilmer, M. E.; Martin, C. D., Linear Algebra Appl. 435 (2011), 641-658.

Third-order tensor product

Let $\mathcal{A} \in \mathbb{F}^{I \times J \times K}$, then

$$\mathrm{unfold}(\mathcal{A}) = \begin{bmatrix} \mathcal{A}^{(1)} \\ \mathcal{A}^{(2)} \\ \vdots \\ \mathcal{A}^{(K)} \end{bmatrix} \in \mathbb{F}^{IK \times J}$$

The $\mathrm{fold}(\cdot)$ is the inverse operation, that is $\mathrm{fold}(\text{unfold}(\mathcal{B})) = \mathcal{B}$.

[1] Kilmer, M. E.; Martin, C. D., Linear Algebra Appl. 435 (2011), 641-658.

t-product[2]

Let $\mathcal{A} \in \mathbb{F}^{I \times J \times K}$ and $\mathcal{B} \in \mathbb{F}^{J \times L \times K}$. Then the *t*-product $\mathcal{A} * \mathcal{B}$ is the tensor in $\mathbb{F}^{I \times L \times K}$ defined as

$$\mathcal{A} * \mathcal{B} = \mathrm{fold}(\mathrm{bcirc}(\mathcal{A})\mathrm{unfold}(\mathcal{B})).$$

[2] Kilmer, M. E.; Martin, C. D., Linear Algebra Appl. 435 (2011), 641-658.

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$$\mathcal{A} * \mathcal{B} = \mathrm{fold}(\mathrm{bcirc}(\mathcal{A})\mathrm{unfold}(\mathcal{B})).$$

- $\mathcal{A} * \mathcal{B} = \mathrm{ifft}(\mathrm{fft}(\mathcal{A}, [\,], 3) \triangle \mathrm{fft}(\mathcal{B}, [\,], 3), [\,], 3).$

[2] Kilmer, M. E.; Martin, C. D., Linear Algebra Appl. 435 (2011), 641-658.

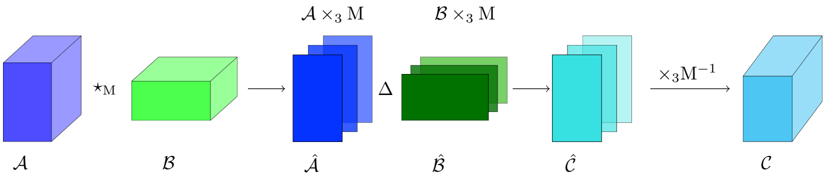
Third-order tensor product

$*$M-product[3]

Let $\mathcal{A} \in \mathbb{C}^{m\times n\times p}$ and $\mathcal{B} \in \mathbb{C}^{n\times \ell\times p}$, and $\mathrm{M} \in \mathbb{C}^{p\times p}$ be an invertible tensor. Then, their $*\mathrm{M}$-product is a tensor $\mathcal{A} \in \mathbb{R}^{m\times \ell\times p}$, and is defined as

$$\mathcal{A} *_{\mathrm{M}} \mathcal{B} = [(\mathcal{A} \times_3 \mathrm{M}) \triangle (\mathcal{B} \times_3 \mathrm{M})] \times_3 \mathrm{M}^{-1},$$

where $\triangle$ denotes the facewise product of tensors.



[3] Kernfeld, E.; Kilmer, M. E.; Aeron, S., Linear Algebra Appl. 485 (2015), 545-570.

Einstein product

Einstein product[4]

The Einstein product, $\mathcal{A} *_N \mathcal{B}$, of tensors $\mathcal{A} \in \mathbb{C}^{I_1\times\cdots\times I_M\times K_1\times\cdots\times K_N}$ and $\mathcal{B} \in \mathbb{C}^{K_1\times\cdots\times K_N\times J_1\times\cdots\times J_L}$ is a tensor in $\mathbb{C}^{I_1\times\cdots\times I_M\times J_1\times\cdots\times J_L}$ and is defined elementwise

$$(\mathcal{A} *_N \mathcal{B})_{i_1\ldots i_M j_1\ldots j_L} = \sum_{k_1,\ldots,k_N=1}^{K_1,\ldots,K_N} a_{i_1\ldots i_M k_1\ldots k_N} b_{k_1\ldots k_N j_1\ldots j_L}.$$

[4]Einstein, A., *The foundation of the general theory of relativity.* In: Kox AJ, Klein MJ, Schulmann R, editors. The collected papers of Albert Einstein 6. Princeton (NJ): Princeton University Press; 2007.p. 146-200.

Step back to the matrix SVD

Singular Value Decomposition

For any $A \in \mathbb{R}^{m \times n}$,

$$A = U \Sigma V^T,$$

where the matrices U and V are orthogonal and the matrix Σ is diagonal with positive real entries.

[4] Lathauwer, L. D.; Moor, B. D.; Vandewalle, J., SIAM J. Matrix Anal. Appl. 21 (2000) 1253-1278.

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k-mode product[5]

Let $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_k \times \cdots \times I_N}$ and $B \in \mathbb{R}^{J \times I_k}$.
Then, their k-mode product is a tensor in $\mathbb{R}^{I_1 \times \cdots \times I_{k-1} \times J \times I_{k+1} \times \cdots \times I_N}$

$$(\mathcal{A} \times_k B)_{i_1 i_2 \ldots i_{k-1} j i_{k+1} \ldots i_N} = \sum_{i_k=1}^{I_k} a_{i_1 i_2 \ldots i_{k-1} i_k i_{k+1} \ldots i_N} b_{j i_k}.$$

[5] Lathauwer, L. D.; Moor, B. D.; Vandewalle, J., SIAM J. Matrix Anal. Appl. 21 (2000) 1253-1278.

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$$\begin{aligned} U \cdot A &= A \times_1 U, \\ A \cdot U^T &= A \times_2 U. \end{aligned}$$

[5] Lathauwer, L. D.; Moor, B. D.; Vandewalle, J., SIAM J. Matrix Anal. Appl. 21 (2000) 1253-1278.

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Singular Value Decomposition

For any $A \in \mathbb{R}^{m \times n}$,

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where the matrices U and V are orthogonal and the matrix Σ is diagonal with positive real entries.

k-mode product[5]

Let $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_k \times \cdots \times I_N}$ and $B \in \mathbb{R}^{J \times I_k}$.
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Mode-n Unfolding

Let $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$. The mode-$n$ unfolding, $\mathcal{A}_{(n)} \in \mathbb{R}^{I_n \times (I_1 \cdot I_2 \cdots I_{n-1} \cdot I_{n+1} \cdots I_N)}$, contains the element $a_{i_1 i_2 \dots i_N}$ at the position with row number i_n and column number equal to

$$
\begin{aligned}
&(i_1 - 1)\, I_2 I_3 \dots I_{n-1} + (i_2 - 1)\, I_3 I_4 \dots I_{n-1} + i_{n-1} + \cdots \\
&+ (i_{n+1} - 1)\, I_1 I_2 \dots I_{n-1} I_{n+2} I_{n+3} \dots I_N \\
&+ (i_{n+2} - 1)\, I_1 I_2 \dots I_{n-1} I_{n+3} I_{n+4} \dots I_N + \cdots + (i_N - 1)\, I_1 I_2 \dots I_{n-1}.
\end{aligned}
$$

Extension to Higher Dimension

Mode-n Unfolding

Let $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$. The mode-$n$ unfolding, $\mathcal{A}_{(n)} \in \mathbb{R}^{I_n \times (I_1 \cdot I_2 \cdots I_{n-1} \cdot I_{n+1} \cdots I_N)}$, contains the element $a_{i_1 i_2 \cdots i_N}$ at the position with row number i_n and column number equal to

$$(i_1 - 1) I_2 I_3 \ldots I_{n-1} + (i_2 - 1) I_3 I_4 \ldots I_{n-1} + i_{n-1} + \cdots$$
$$+ (i_{n+1} - 1) I_1 I_2 \ldots I_{n-1} I_{n+2} I_{n+3} \ldots I_N$$
$$+ (i_{n+2} - 1) I_1 I_2 \ldots I_{n-1} I_{n+3} I_{n+4} \ldots I_N + \cdots + (i_N - 1) I_1 I_2 \ldots I_{n-1}.$$

Example

Define a tensor $\mathcal{A} \in \mathbb{R}^{3 \times 2 \times 3}$ by $a_{111} = a_{112} = a_{211} = -a_{212} = 1$, $a_{213} = a_{311} = a_{313} = a_{121} = a_{122} = a_{221} = -a_{222} = 2$, $a_{223} = a_{321} = a_{323} = 4$, $a_{113} = a_{312} = a_{123} = a_{322} = 0$. The matrix unfolding $\mathbf{A}_{(1)}$ is given by

$$\mathbf{A}_{(1)} = \left(\begin{array}{rrr|rrr} 1 & 1 & 0 & 2 & 2 & 0 \\ 1 & -1 & 2 & 2 & -2 & 4 \\ 2 & 0 & 2 & 4 & 0 & 4 \end{array} \right).$$

Frame Title

Higher-order Singular Value Decomposition[6]

Let $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$. Then,

$$\mathcal{A} = \mathcal{S} \times_1 \mathbf{U}^{(1)} \times_2 \mathbf{U}^{(2)} \times_3 \cdots \times_N \mathbf{U}^{(N)},$$

- where $\mathbf{U^{(n)}}$ is determined from the SVD of mode-n unfolding, $\mathcal{A}_{(n)}$, of tensor $\mathcal{A}$, say,

$$\mathcal{A}_{\mathbf{(n)}} = \mathbf{U^{(n)}} \cdot \mathbf{\Sigma^{(n)}} \cdot \mathbf{V^{(n)}}^T.$$

- Compute $\mathcal{S} = \mathcal{A} \times_1 \mathbf{U^{(1)}}^T \times_2 \mathbf{U^{(2)}}^T \times_3 \cdots \times_N \mathbf{U^{(N)}}^T$.

[6] Lathauwer, L. D.; Moor, B. D.; Vandewalle, J., SIAM J. Matrix Anal. Appl. 21 (2000) 1253-1278.

Extension to Higher Dimension

$*$M-product

Let $\mathcal{A} \in \mathbb{R}^{m\times n\times p}$, $\mathcal{B} \in \mathbb{R}^{n\times k\times p}$ and $M \in \mathbb{R}^{p\times p}$. Then the $*$M-product of tensors $\mathcal{A}$ and $\mathcal{B}$ is denoted by $\mathcal{A} *_M \mathcal{B} \in \mathbb{R}^{m\times k\times p}$ and defined as

$$\mathcal{A} *_M \mathcal{B} = [(\mathcal{A} \times_3 M)\Delta(\mathcal{B} \times_3 M)] \times_3 M^{-1}.$$

[6] Kernfeld, E.; Kilmer, M.; Aeron, S., Linear Algebra Appl. 485 (2015), 545-570.

Extension to Higher Dimension

Tensor transpose[7]

Let $\mathcal{A} \in \mathbb{R}^{m \times n \times p}$ and $M \in \mathbb{R}^{p \times p}$ be an invertible matrix. Then its transpose under $\star_{\mathrm{M}}$ product is denoted by $\mathcal{A}^T$ and is defined as

$$\mathcal{A}^T(:,:,i) = \left(\widehat{\mathcal{A}}^{\mathrm{T}} \times_3 M^{-1}\right)(:,:,i), \quad i = 1, \ldots, p.$$

$*$M-product

Let $\mathcal{A} \in \mathbb{R}^{m \times n \times p}$, $\mathcal{B} \in \mathbb{R}^{n \times k \times p}$ and $M \in \mathbb{R}^{p \times p}$. Then the $*$M-product of tensors $\mathcal{A}$ and $\mathcal{B}$ is denoted by $\mathcal{A} *_M \mathcal{B} \in \mathbb{R}^{m \times k \times p}$ and defined as

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[7]Kernfeld, E.; Kilmer, M.; Aeron, S., Linear Algebra Appl. 485 (2015), 545-570.

Extension to Higher Dimension

SVD in the framework of $*$M-product[8]

Let $\mathcal{A} \in \mathbb{R}^{m \times n \times p}$ and $M \in \mathbb{R}^{p \times p}$ be an invertible matrix. Then, there exist orthogonal tensors $\mathcal{U} \in \mathbb{R}^{m \times m \times p}$ and $\mathcal{V} \in \mathbb{R}^{n \times n \times p}$ such that

$$\mathcal{A} = \mathcal{U} *_M \Sigma *_M \mathcal{V}^T,$$

where $\Sigma \in \mathbb{R}^{m \times n \times p}$ is F-diagonal tensor.

[8] Kilmer, M. E.; Horesh, L.; Avron, H.; Newman, E., Proceedings of the National Academy of Sciences, 118 (2021), e2015851118.

Extension to Higher Dimension

Einstein product

Let $\mathcal{A} \in \mathbb{R}^{I_1 \times \cdots \times I_M \times J_1 \times \cdots \times J_N}$ and $\mathcal{B} \in \mathbb{R}^{J_1 \times \cdots \times J_N \times K_1 \times \cdots \times K_L}$. Then,

$$(\mathcal{A} *_N \mathcal{B})_{i_1 i_2 \ldots i_M k_1 k_2 \ldots k_L} = \sum_{j_1, j_2, \ldots, j_N = 1}^{J_1, J_2, \ldots, J_N} a_{i_1 i_2 \ldots i_M j_1 j_2 \ldots j_N} b_{j_1 j_2 \ldots j_N k_1 k_2 \ldots k_L}.$$

Extension to Higher Dimension

Tensor transpose[9]

Let $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M \times J_1 \times J_2 \times \cdots \times J_N}$. Then the transpose of the tensor $\mathcal{A}$, denoted by $\mathcal{A}^T$, is a tensor in $\mathbb{R}^{J_1 \times J_2 \times \cdots \times J_N \times I_1 \times I_2 \times \cdots \times I_M}$ and defined elementwise,

$$\mathcal{A}^T_{j_1 j_2 \ldots j_N i_1 i_2 \ldots i_M} = \mathcal{A}_{i_1 i_2 \ldots i_M j_1 j_2 \ldots j_N}.$$

Einstein product

Let $\mathcal{A} \in \mathbb{R}^{I_1 \times \cdots \times I_M \times J_1 \times \cdots \times J_N}$ and $\mathcal{B} \in \mathbb{R}^{J_1 \times \cdots \times J_N \times K_1 \times \cdots \times K_L}$. Then,

$$(\mathcal{A} *_N \mathcal{B})_{i_1 i_2 \ldots i_M k_1 k_2 \ldots k_L} = \sum_{j_1, j_2, \ldots, j_N = 1}^{J_1, J_2, \ldots, J_N} a_{i_1 i_2 \ldots i_M j_1 j_2 \ldots j_N} b_{j_1 j_2 \ldots j_N k_1 k_2 \ldots k_L}.$$

[9] Sun, L.; Zheng, B.; Bu, C.; Wei, Y., Linear Multilinear Algebra 64 (2016), 686-698

Extension to Higher Dimension

SVD framed in Einstein product[10]

For a tensor $\mathcal{A} \in \mathbb{R}^{I_1 \times \cdots \times I_N \times J_1 \times \cdots \times J_N}$, the *SVD* of $\mathcal{A}$ has the form as

$$\mathcal{A} = \mathcal{U} *_N \mathcal{B} *_N \mathcal{V}^T,$$

where $\mathcal{U} \in \mathbb{R}^{I_1 \times \cdots \times I_N \times I_1 \times \cdots \times I_N}$ and $\mathcal{V} \in \mathbb{R}^{J_1 \times \cdots \times J_N \times J_1 \times \cdots \times J_N}$ are orthogonal tensors, $\mathcal{B} \in \mathbb{R}^{I_1 \times \cdots \times I_N \times J_1 \times \cdots \times J_N}$ is a diagonal tensor.

[10] Sun, L.; Zheng, B.; Bu, C.; Wei, Y., Linear Multilinear Algebra 64 (2016), 686-698

Einstein product

Moore–Penrose inverse[11] (Einstein product)

Let $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N \times J_1 \times J_2 \times \cdots \times J_N}$. Then there exists a unique tensor $\mathcal{X} \in \mathbb{R}^{J_1 \times J_2 \times \cdots \times J_N \times I_1 \times I_2 \times \cdots \times I_N}$ such that

$$
\begin{aligned}
\mathcal{A} *_N \mathcal{X} *_N \mathcal{A} &= \mathcal{A}; \\
\mathcal{X} *_N \mathcal{A} *_N \mathcal{X} &= \mathcal{X}; \\
(\mathcal{A} *_N \mathcal{X})^T &= \mathcal{A} *_N \mathcal{X}; \\
(\mathcal{X} *_N \mathcal{A})^T &= \mathcal{X} *_N \mathcal{A}.
\end{aligned}
$$

[11] Sun, L.; Zheng, B.; Bu, C.; Wei, Y., Linear Multilinear Algebra 64 (2016), 686-698

Thank You!